ADVANCED PROBLEMS FOR LECTURE 20

In the lecture, we will be talking at a more general level of the patterns of what is going on. But there are deeper questions to explore therein, which we cannot get to. This is your opportunity to investigate deeper for yourself. Don't limit yourself to just these questions, let your curiosity guide you too! That is, if you wonder about any details, or if any certain patterns or ideas exist, see if you can state your question clearly, and answer it! Here are a few suggestions to get you started.

(1) Is it possible for the area to exist between two non-integrable functions?

(2) Periodicity is a kind of symmetry that can be exploited with integrals. Suppose that f is a T-periodic continuous function on $\mathbb{R}$. That is, $f(x+T) = f(x)$ for all $x \in \mathbb{R}$.
   (a) Show that f is bounded. That is, f attains an absolute maximum and absolute minimum.
   (b) Show that
   $$\int_x^{x+T} f(t)dt = \int_0^T f(t)dt$$
   That is, there is some real number I, which is the common integral of $f(x)$ over any period.
   (c) Show that for any $a < b$, we have
   $$\int_a^b f(t)dt = \lfloor \frac{b-a}{T} \rfloor I + \int_0^{b-a-\lfloor \frac{b-a}{T} \rfloor T} f(t)dt$$
   (d) As an example, we know that $\sin(x)$ is 2π-periodic. So show, using the above result that for any $a < b$
   $$\int_a^b \sin(x)dx = \int_0^{b-a-\lfloor \frac{b-a}{2\pi} \rfloor T} \sin(x)dx$$

(3) Show that the integral is translation invariant. That is, show that for any $c \in \mathbb{R}$, we have
$$\int_a^b f(x)dx = \int_{a+c}^{b+c} f(x-c)dx$$

Translation invariant integrals like what we described in our discussion of exploiting the symmetry of periodicity, are known as "Haar measures". The usual integral also exhibits a translation invariance too, as we see in the problems that follow our periodicity discussion. It might be fun to think about how our calculation of volumes of solids of revolution are exploiting these symmetries to reduce our integration efforts to a one-dimensional integral.